

Scope of Mass Derivations in FFGFT: Rest Masses, Hadron Dynamics, and Open Bridges

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Starting point: $E = m_0 c^2$ is the exact rest energy ($p = 0$), not an approximation; for small v , $E \approx m_0 c^2 + \frac{1}{2} m_0 v^2$.
- Proton mass: row "Gluon kinetic energy" $\rightarrow$ QCD energy (quark kinetic energy, gluon field energy, trace anomaly).
- Neutrino table: the rows come from Doc. 006 (not 007) and are not viable according to Doc. 190 R109 (note).
- $v \cdot \xi^{1/3} \approx 200 \text{ MeV} \rightarrow \approx 12.6 \text{ GeV}$ (does not match Λ_{QCD}); $\xi^{-1/3} = 9.65 \rightarrow 19.57$.
- Second table (1 Oct. 2026): "gluon kinetic energy" $\rightarrow$ QCD energy (quark kinetic energy, gluon field energy, trace anomaly).

List of symbols

Symbol	Meaning	Value / unit
M_Z	Mass of the Z boson; reference scale for α_s	91.19 GeV
Λ_{QCD}	QCD confinement scale	$\approx 200\text{--}217$ MeV
$\alpha_s(\mu)$	Strong coupling at scale μ	$\alpha_s(M_Z) \approx 0.118$; $\alpha_s(m_\tau) \approx 0.33$
$D = N_c$	Number of colour degrees of freedom (Doc. 160)	$D = 3$
y_e	Yukawa coupling of the electron	$m_e/v \approx 2.08 \cdot 10^{-6}$
v	Higgs VEV	246 GeV
m_p, m_n	Proton / neutron mass	938.3 / 939.6 MeV
RGE	Renormalisation group equation	describes μ -dependence of α_s
K_{frak}	Fractal interface factor (rolled-up $\rightarrow$ SI)	$1 - 100\xi \approx 0.987$
$m^{\text{bare}} / m^{\text{SI}}$	Geometric / measured mass	nat. units / MeV
ξ	FFGFT geometry parameter	$4/30000 \approx 1.333 \cdot 10^{-4}$

Summary

The FFGFT derives particle masses from the geometry parameter $\xi = 4/30000$ on the compact 4D torus $T^4/\mathbb{Z}_3$. This scope is precise for rest masses from the Higgs-Yukawa sector (leptons, quark rest masses). Hadron masses such as the proton mass contain a dominant non-perturbative QCD contribution that requires its own framework. This document draws the boundary explicitly and grounds it physically in the relativistic energy relation $E^2 = (m_0c^2)^2 + (pc)^2$ and the picture of massless gluons inside the proton.

Status markers

[K] FFGFT derives rest masses (leptons, quark Yukawa) from ξ -geometry. **Hadron masses** have additionally a non-perturbative QCD contribution: **open bridge**.

1 Starting point: the complete energy relation

The popular $E = m_0c^2$ is the rest energy, i.e. the special case $p = 0$ of the complete relation below; as such it is exact. Linearising the Lorentz factor

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx 1 + \frac{1}{2} \frac{v^2}{c^2} \quad (v \ll c) \quad (1)$$

gives, for small velocities, $E = \gamma m_0c^2 \approx m_0c^2 + \frac{1}{2}m_0v^2$. (Corrected on 1 Oct. 2026: previously "The popular $E = m_0c^2$ is an approximation. It follows from linearising the Lorentz factor ... and holds only at small velocities" – $E(p = 0) = m_0c^2$ is exact; $\gamma m_0c^2 = m_0c^2 + \frac{1}{2}m_0v^2 + O(v^4)$.) The complete, exact relation is:

$$\boxed{E^2 = (m_0c^2)^2 + (pc)^2} \quad (2)$$

This equation is the actual core – not the simplified formula. The second term pc does not vanish when a particle is massless. For massless particles ($m_0 = 0$):

$$E = pc, \quad (3)$$

so a massless particle moving at the speed of light still contributes to the total energy – and thus to the measurable mass of a system.

.2 The proton: mass as emergent property

The proton mass $m_p \approx 938.3$ MeV is the clearest example of this emergence:

Table 1: Energy contributions to the proton mass

Contribution	Value	Fraction	Mechanism
u and d quark rest masses	≈ 10 MeV	$\approx 1\%$	Higgs-Yukawa coupling
QCD energy (quark kinetic energy, gluon field energy, trace anomaly)	≈ 928 MeV	$\approx 99\%$	QCD confinement
Proton mass	938.3 MeV	100 %	

(Corrected on 1 Oct. 2026: previously row “Gluon kinetic energy” – according to the lattice QCD decomposition (Yang et al., PRL 121, 212001 (2018)) the proton mass splits into quark kinetic energy ($\sim 32\%$), gluon field energy ($\sim 36\%$), trace anomaly ($\sim 23\%$) and quark mass term ($\sim 9\%$); the $\approx 1\%$ row counts only the valence current-quark masses.)

The proton consists of three quarks (uud), whose rest masses together amount to about 10 MeV. The remainder – 928 MeV, around 99 % – is QCD energy: kinetic energy of the quarks, field energy of the gluons, the carriers of the strong force, and the trace anomaly. (Corrected on 1 Oct. 2026: previously “comes from gluons” – decomposition see the note below Table 1.) Gluons are massless ($m_0 = 0$). They move at the speed of light. By Eq. (2) they nevertheless contribute to the total energy of the system through their momentum.

Timelessness inside the proton

For massless particles moving at the speed of light, time dilation is maximal: the proper time of these particles is zero. Gluons inside the proton exist in a state where no time passes. For them, the spatial extent of the world is zero – length contraction is maximal.

And yet: these timeless, massless constituents account for the dominant part of what is measured externally as the proton mass. Mass is a property at the surface of a system, not an intrinsic property of its components.

[S] This observation connects directly to the FFGFT fundamental relation. $\tilde{T} \cdot m = 1$ states: intrinsic time and mass are dual. The interior of the proton is precisely the timeless level – and this constitutes the majority of the externally apparent mass. Rest mass arises through projection from the timeless to the time-bearing domain. The proton illustrates $\tilde{T} \cdot m = 1$ inadvertently but precisely. This connection is conceptually robust but not yet a formal derivation; it is noted for a future document.

.3 Two sectors – two mechanisms

The ratio $m_p/m_e \approx 1836$ connects two physically distinct sectors:

Quantity	Sector	Scale
m_e, m_μ, m_τ	Higgs-Yukawa	$\xi^{P_i} \cdot r_i \cdot v$
$m_{\nu_e}, m_{\nu_\mu}, m_{\nu_\tau}$	Higgs-Yukawa (suppressed)	$\xi^{Q_i} \cdot v$
m_u, m_d, m_s (rest)	Higgs-Yukawa	(open, Doc. 006)
m_p, m_n (total)	QCD confinement + Higgs-Yukawa	$\Lambda_{\text{QCD}} + m_{\text{quarks}}$

[B] Proton mass and electron mass arise through entirely different physics. m_e is a rest mass from the Higgs-Yukawa sector; m_p consists to 99 % of QCD confinement energy. A ratio of these two quantities connects two independent scales (Λ_{QCD} and $y_e \cdot v$). There is no structural reason why the geometry of a single parameter ξ should simultaneously fix both scales.

Consequence: m_p/m_e is not a derivation target of the current FFGFT corpus. An apparently exact five-digit agreement would be a warning signal, not a success criterion – because QCD contributions carry percentage-level theory uncertainties.

.4 Neutrinos: the other limiting case

In contrast to the proton, neutrinos are the opposite limit: extremely small rest masses lying entirely within the Higgs-Yukawa sector.

[S] In FFGFT, neutrinos are treated as “nearly massless photons” (Doc. 007): they carry the same (n_θ, n_ϕ) winding structure as the charged leptons, but with double ξ -suppression. This yields rest masses in the sub-eV range, consistent with cosmological bounds ($\sum m_\nu < 0.12$ eV, Planck 2018).

Table 2: Neutrino quantum numbers and mass suppression (Doc. 007)

Neutrino	n_θ	n_ϕ	p_i	Factor	Status
ν_e	3	2	5/2	$\xi^{5/2}$	[S]
ν_μ	5	4	3	ξ^3	[S]
ν_τ	9	5	25/9	$\xi^{25/9}$	[S]

(Note of 1 Oct. 2026: The neutrino rows come from Doc. 006 (not 007); according to Doc. 190 R109 they are not viable – $p_{\nu_\tau} = 25/9$ is the prefactor $r_{\tau\tau}$ copied into the exponent column. The authoritative treatment of neutrinos is Doc. 340.)

Neutrinos demonstrate that the Higgs-Yukawa sector admits unlimited suppression through powers of ξ . The proton lies outside this sector.

.5 Rolled up and unrolled: the structural reason for the two-sector separation

Doc. 311 distinguishes three ways in which the fourth spatial direction of the 4D torus T^4 becomes the three-dimensional world:

Way	Mechanism	Problem
Compactification (Kaluza-Klein)	fourth direction becomes a circle, windings = modes	radius R is a free modulus
Projection (cut-and-project)	4D lattice $\rightarrow$ 3D cut	loses lattice invariants
Fourth is not space	carries mass, time, internal index	without compactness no mode tower

[K] FFGFT combines ways 1 and 3, bound together by $\tilde{T} \cdot m = 1$: the fourth direction is *rolled up* (compact) *and bound* (it carries mass). The radius R is therefore not free – it is the mass. This is the only case in which a scale arises without bringing in an additional parameter.

Rolled up: rest masses from the Yukawa sector

[K] Quark and lepton rest masses arise from *rolled-up* modes: winding numbers (n_θ, n_ϕ) on the compact torus generate quantised masses $m_i = r_i \cdot \xi^{p_i} \cdot v$ (Docs. 006, 317). Integer winding numbers are forced by the compactness of the torus. This is the geometric origin of the Yukawa couplings: not free parameters, but topological invariants.

Unrolled: gluons as dynamic fields in 3D space

[S] Gluons are massless vector bosons propagating in the unrolled, non-compact 3D space. They are not winding modes on the torus but dynamic degrees of freedom of the gauge field in the unfolded space. Their energy does not come from ξ -windings but from motion and QCD confinement.

The proton in this language:

Contribution	Origin	Sector
Quark rest masses (~ 1 %)	rolled-up modes, Yukawa	FFGFT, [K]
QCD energy (~ 99 %: quark kinetic energy, gluon field energy, trace anomaly)	unrolled modes, QCD dynamics	open bridge, [S]

Hadron masses have contributions from *both* regimes – that is the structural reason why a unified ξ -derivation does not directly apply. The boundary between rolled-up and unrolled is the boundary of the scope.

.6 The fractal correction K_{frak} : interface factor in the transition rolled-up $\rightarrow$ unrolled

[K] K_{frak} is the factor that appears in the transition from the geometrically rolled-up (bare) description to the measured (unrolled, SI) description (Docs. 012, 133):

$$m_i^{\text{SI}} = K_{\text{frak}} \cdot C_{\text{conv}} \cdot m_i^{\text{bare}}, \quad K_{\text{frak}} = 1 - 100\xi \approx 0.9867 \quad (4)$$

The physical justification is the fractal porosity of spacetime at Planck scales ($D_{\text{frak}} = 2.94$, Doc. 003), which reduces the effective interaction strength relative to a smooth space.

The key point: K_{frak} cancels in ratios

[K] In mass ratios K_{frak} drops out completely (Doc. 012):

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu^{\text{bare}}}{K_{\text{frak}} \cdot m_e^{\text{bare}}} = \frac{m_\mu^{\text{bare}}}{m_e^{\text{bare}}} \quad (5)$$

This is structurally significant: the mass ratios that FFGFT derives in the lepton sector (m_μ/m_e , m_τ/m_μ , Doc. 317) are purely geometric and do *not* require K_{frak} . K_{frak} appears only when absolute SI values are needed – for the gravitational constant G , the fine-structure constant α , and absolute mass values in MeV or kg.

Scope and limitation for hadrons

K_{frak} is the interface factor between two levels:

Level	Description	K_{frak}
Rolled-up (bare)	m^{bare} , torus geometry, ξ -powers	not needed
Transition	$m^{\text{SI}} = K_{\text{frak}} \cdot C_{\text{conv}} \cdot m^{\text{bare}}$	here
Unrolled (SI)	measurable mass in MeV, kg	already included

For the **hadron sector**, K_{frak} would appear in the transition $\Lambda_{\text{QCD}}^{\text{bare}} \rightarrow \Lambda_{\text{QCD}}^{\text{SI}}$ – but $\Lambda_{\text{QCD}}^{\text{bare}}$ itself is not derived from ξ . K_{frak} as an interface factor does not resolve the underlying problem: the 928 MeV contribution is missing not because of a lacking conversion, but because the underlying QCD mechanism is not yet treated in the FFGFT corpus.

A135: settability as a diagnostic

[K] A135 states the settability criterion: what can be norm-set to one was a property of the measurement system. $K_{\text{frak}} = 1 - 100\xi$ cannot be set to 1 without changing ξ – it is bound to something physical, not a pure measurement-system artefact. For precision predictions Doc. 133 applies.

.7 Scope of the ξ mass derivations

Table 3: Complete scope of mass derivations in FFGFT

Particle	Dominant contribution	Status	Doc.
e, μ, τ	Higgs-Yukawa, rest mass	[K]	006, 317
ν_e, ν_μ, ν_τ	Higgs-Yukawa, ξ^2 -suppressed	[S]	007
u, d, s, c, b, t (rest)	Higgs-Yukawa	[S] open	006
W, Z	Higgs mechanism	[S] scale correct	—
Higgs h	Higgs self-coupling	[K] ξ_{Higgs}	A-series
p, n (total)	QCD + Higgs-Yukawa	open bridge	—
m_p/m_e	sector-mixed	not claimed	—

.8 Three scenarios for the QCD bridge

Whether ξ also captures the QCD sector is an open research question. Three scenarios are conceivable:

1. **ξ fixes Λ_{QCD} via orbifold geometry.** The $T^4/\mathbb{Z}_3$ structure generates nine orbifold fixed points. Whether these force a natural QCD scale is open. Doc. 041 lists the ansatz $\Lambda_{\text{QCD}} = v \cdot \xi^{1/3}$; evaluated, it gives $v \cdot \xi^{1/3} \approx 12.6$ GeV and misses $\Lambda_{\text{QCD}} \sim 217$ MeV (factor ≈ 58 ; correction in Doc. 041 of 30 Sept. 2026). (*Corrected on 1 Oct. 2026: previously “ ≈ 200 MeV ... This is a rough match” – $246\,000 \text{ MeV} \cdot \xi^{1/3} = 246\,000 \cdot 0.05109 \text{ MeV} = 12\,568 \text{ MeV}$.) A derivation of the QCD scale from ξ is therefore not available. **[S]***
2. **Trace anomaly on compact spaces.** The renormalisation group for the strong coupling $\alpha_s(\mu)$ on compact tori generates a natural infrared scale. Khanna et al. (2014, arXiv:1409.1245) provide the mathematical framework; whether ξ enters there is unexplored. **[S]**

3. Hadron sector with SM inputs (Doc. 005) and geometric α_s derivation (Doc. 160).

Doc. 005 contains a complete proton calculation (0.938100 GeV, deviation 0.02 %) using the extended fractal formula. It uses $\alpha_s = 0.118$ and $\Lambda_{\text{QCD}} = 0.217$ GeV as Standard Model inputs. The claim “without free parameters” means: no additional parameters beyond the SM inputs.

Essential precision: Doc. 160 (lepton lifetimes) derives α_s geometrically:

$$\alpha_s(m_\tau) = D \cdot \xi^{1/(D+1)} = 3 \cdot \xi^{1/4} \approx 0.322 \quad (\text{exp: } 0.330, -2.3\%) \quad (6)$$

$D = 3$ colour degrees of freedom, $D + 1 = 4$ spacetime dimensions. This is α_s at scale m_τ , not at M_Z . The strong coupling runs with energy: $\alpha_s(m_\tau) \approx 0.33$ and $\alpha_s(M_Z) \approx 0.118$ are different values of the *same* running coupling. The FFGFT derivation targets the correct scale (Doc. 160) and uses 0.118 for the proton calculation in Doc. 005 because M_Z is the relevant scale there. **[K]**

Open point: the complete RGE bridge from $\alpha_s(m_\tau)$ to $\alpha_s(M_Z)$ within FFGFT is not yet carried out. **[S]**

Scenario 3 is the current position. Scenarios 1 and 2 are marked research questions. The approach in Doc. 041 ($\Lambda_{\text{QCD}} = v \cdot \xi^{1/3}$) does not support Scenario 1, since it yields ≈ 12.6 GeV instead of ~ 217 MeV. *(Corrected on 1 Oct. 2026: previously “is a first step towards Scenario 1 but not yet a complete derivation” – $v\xi^{1/3} = 12.57$ GeV, see Scenario 1.)*

.9 Relation to the corpus

Structural classification of the early documents

[K] The early documents (Docs. 001–041) primarily worked out the *structural connections* between ξ and the physical constants – not precision derivations. The formulas in these documents are order-of-magnitude approaches: they show *that* and *how* a coupling constant depends on ξ , not necessarily the exact prefactor.

Document	Content	Character
Doc. 041	$\alpha_s = \xi^{-1/3}$, $\alpha_W = \xi^{1/2}$ (nat. units)	structural sketch [S]
Doc. 041	$\Lambda_{\text{QCD}} = v \cdot \xi^{1/3} \approx 12.6$ GeV	misses Λ_{QCD} [S]
Doc. 160	$\alpha_s(m_\tau) = 3\xi^{1/4} \approx 0.322$	precision derivation [K]
Doc. 011	α from torus geometry	precision derivation [K]
Doc. 006	lepton masses $m_i = r_i \xi^{p_i} v$	precision derivation [K]

(Corrected on 1 Oct. 2026: previously table entry “ ≈ 200 MeV, order of magnitude” – $v\xi^{1/3} = 12.57$ GeV, a factor ≈ 63 above 200 MeV; cf. Doc. 041.)

In particular: Doc. 041 lists $\alpha_s \approx 0.118$ (experiment) and $\xi^{-1/3} = 19.57$ (natural units) *(Corrected on 1 Oct. 2026: previously $9.65 - \xi^{-1/3} = 7500^{1/3} = 19.574$; corrected likewise in Doc. 041.)* side by side without unit conversion. This is not an error in the physics but a missing bridge – the conversion from natural to SI units is absent there. Doc. 160 provides the cleaner derivation.

Corpus references

- **Doc. 005** (tm-extension): proton calculation with extended fractal formula; $\alpha_s = 0.118$ and $\Lambda_{\text{QCD}} = 0.217$ GeV as SM inputs (not ξ -derivations).
- **Doc. 006** (particle masses): lepton rest masses **[K]**; hadron masses not treated.

- **Doc. 007** (neutrinos): double ξ -suppression, quantum number assignment, cosmological bounds.
- **Doc. 041** (parameter derivation): structural connections of all couplings; order-of-magnitude estimates, not SI precision.
- **Doc. 160** (lepton lifetimes): geometric derivation $\alpha_s(m_\tau) = 3\xi^{1/4}$, precision -2.3% [K].
- **Doc. 317** (KSAU-FFGFT): complete quantum number table.
- **A-series:** ξ_{Higgs} , Koide relation.

.10 Entry in Doc. 190 (R76)

This clarification is declared as register entry R76 in Doc. 190: scope of mass derivations explicitly delimited; hadron sector documented as open bridge; m_p/m_e not claimed as a derivation target of the current corpus; three scenarios for the QCD bridge marked as research questions.